



Predictive simulations of pathological gait: from clinical data to digital twins

Tom Buurke, Friedl De Groote, Ines Vandekerckhove, Tim van der Zee, Ellis Van Can, Míriam Febrer Nafria, Lars D'Hondt, Stefanie de Jager

Workshop W1 ESMAC 2026, Maastricht, NL

umcg:

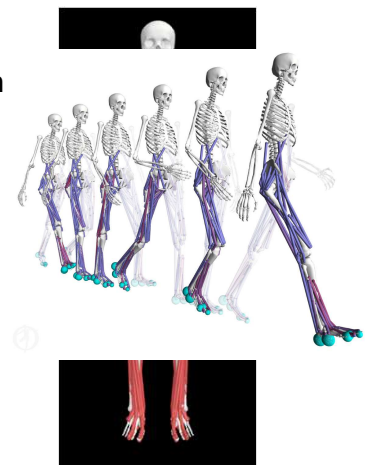
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1

Predictive simulations for clinical gait analysis

- What is a musculoskeletal model?
 - Mathematical description of musculoskeletal system
- Segments: Rigid bodies
- Idealized joints
- Muscles: Line segments
 - Actuators that exert force along a line of action
- Muscles generate force and movement



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PredSim

2

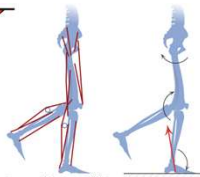
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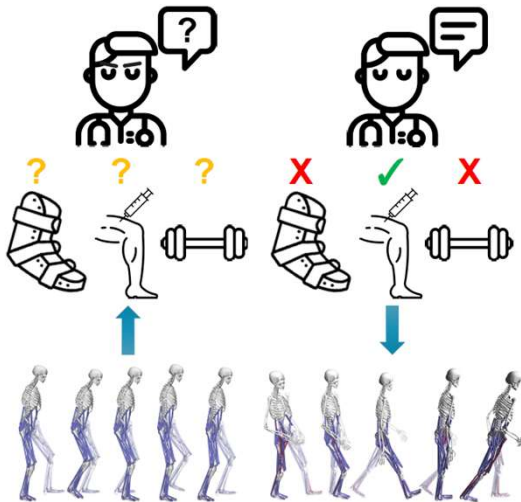
Why do we use musculoskeletal models?

From recorded data

- Model what we cannot measure
 - Muscle force
 - Internal load
 - Muscle-tendon length

Predict movement

- 
- 
- 
- Assistive devices



Illustrations from Uchida & Delp (2020)

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3

3

How do we use musculoskeletal models?

Open-source software

- OpenSim
- PredSim
- SCONE
- Moco
- MyoSuite
- ...

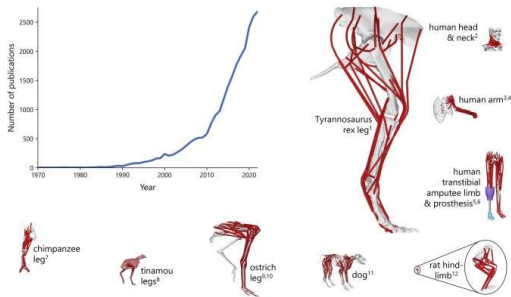
Commercial software

- Anybody
- HyFyDy
- ...



Simulation CPU time

- 2001: 10,000 hours¹
- 2019: 1 hour²



1) Anderson & Pandy (2001)
2) Falisse et al. (2019)

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PredSim

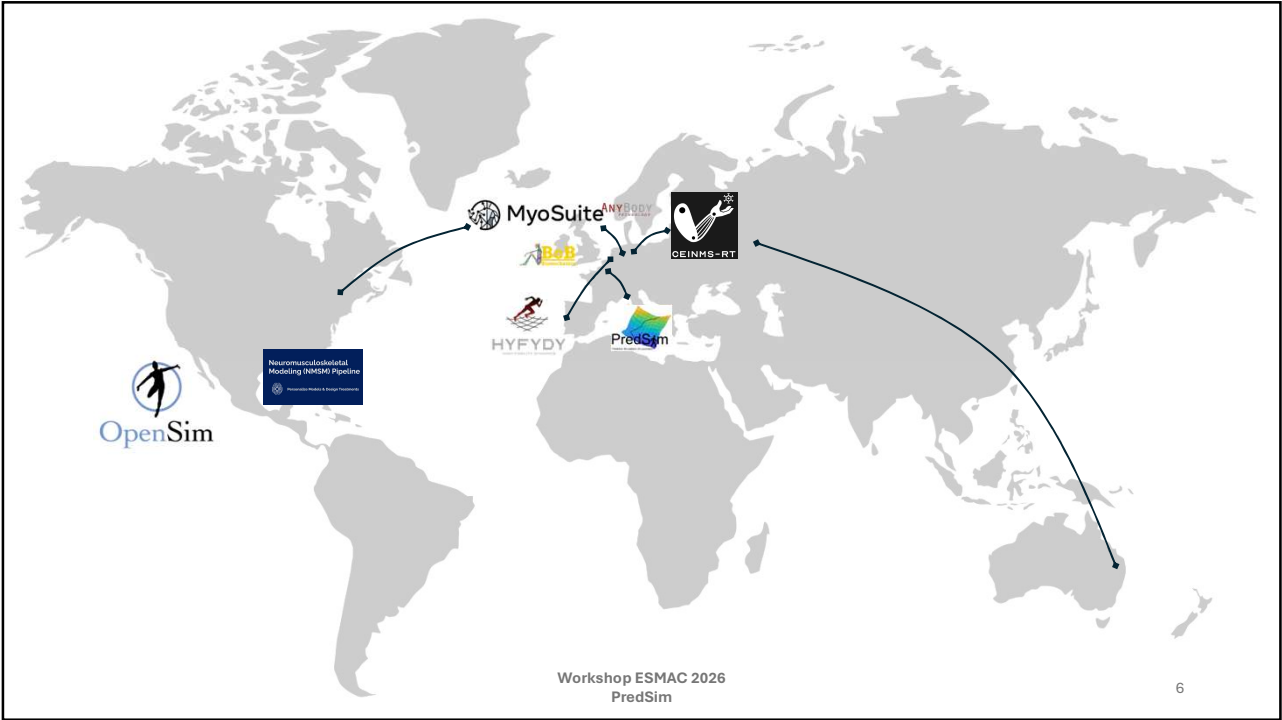
Figure from Uhlrich et al. (2023)

4

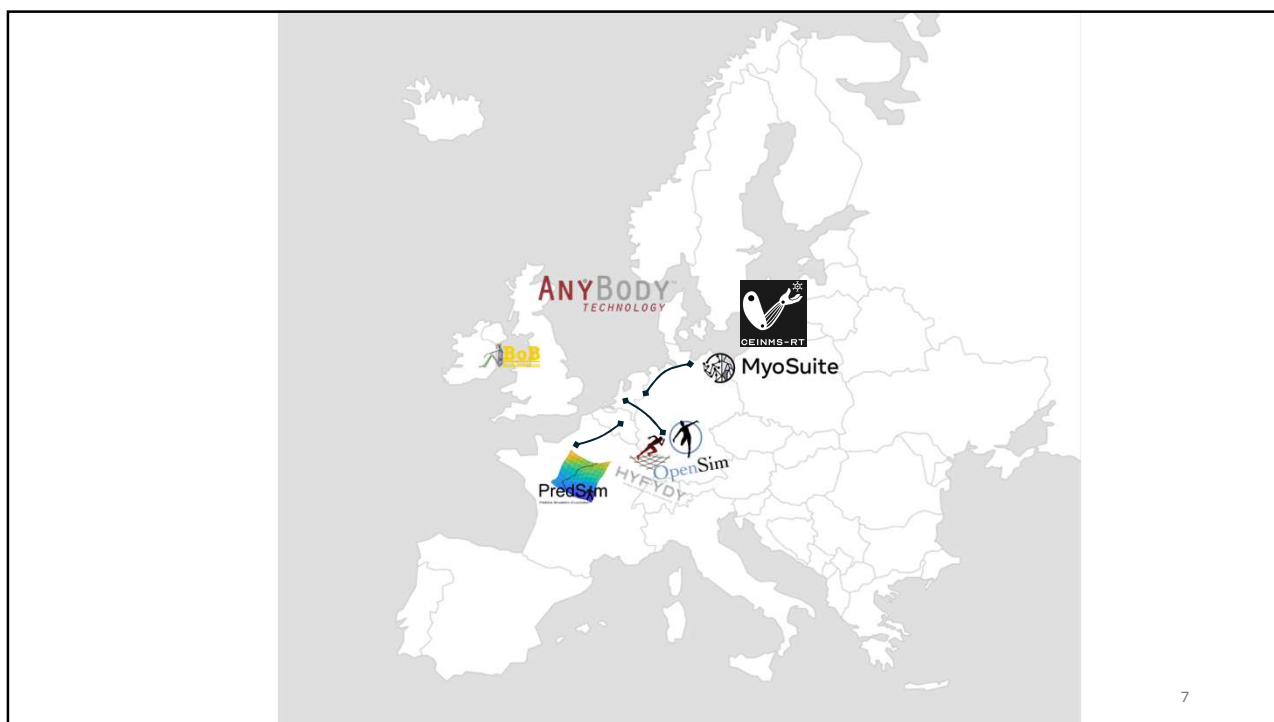
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5



6



7

Our goals for today

After this workshop you...

- will know the theory underlying musculoskeletal modelling and predictive simulations.
- can perform predictive simulations with 2D models in *PredSim*.
- can find information and sources for 3D simulations.
- will know the opportunities and barriers for clinical implementation of predictive simulations.



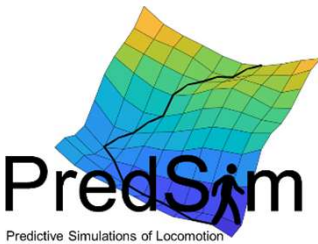
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PredSim

8

8

Program

- 08:30 – 08:50 Introduction and theory
- Predictive simulations for clinical gait analysis. *Tom Buurke*
 - Underlying theory of modelling and simulations with *PredSim*. *Friedl De Groote*
- 08:50 – 09:10 Today’s clinical cases: What can we personalize, model and simulate?
- Muscle weakness and -stiffness in Duchenne muscular dystrophy. *Ines Vandekerckhove*
 - Surgical intervention in cerebral palsy. *Ellis Van Can*
 - The effect of an ankle-foot-orthosis in foot drop. *Tim van der Zee*
- 09:10 – 11:30 Hands-on: Predictive simulations of the 3 clinical cases in 2D with *PredSim*
- Refreshment break during hands-on session.
- 11:30 – 12:00 Closing discussion
- Showcase: Predictive simulations in 3D. *Friedl De Groote*
 - Group discussion: Opportunities and barriers for clinical implementation of predictive simulations. *Tom Buurke*

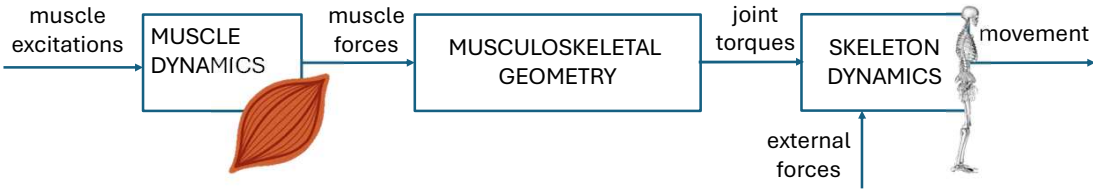


9

Simulations based on a musculoskeletal model



dynamic constraints

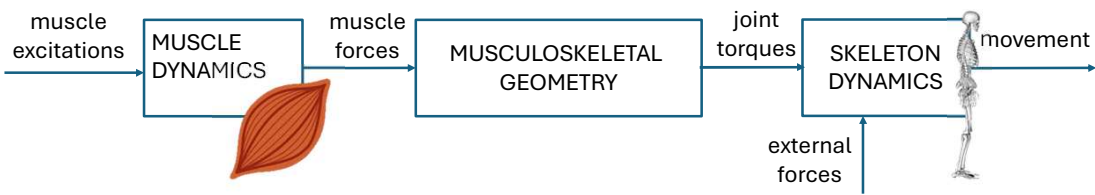


task constraints, e.g. average forward speed and periodicity for locomotion

10

Optimal control assumption

find muscle excitations **that**
minimize cost
subject to
dynamic constraints



task constraints, e.g. average forward speed and periodicity for locomotion

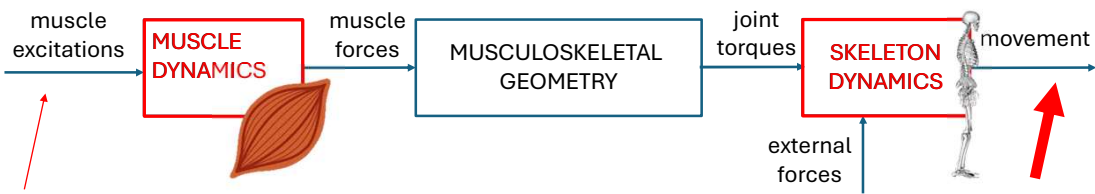
11

11

Solving optimal control problems

find muscle excitations **that**
minimize cost
subject to
dynamic constraints

STIFF DIFFERENTIAL EQUATIONS



task constraints, e.g. average forward speed and periodicity for locomotion

12

12

We reduced computational cost by factor 20

Challenge: High computational cost of numerical methods

Muscle-driven 3D model (23 dofs, 54 muscles)
~10,000 hours (Anderson and Pandy, 2001)

2001

Muscle-driven 3D model (23 dofs, 40 muscles)
~60 hours (Miller, 2014)

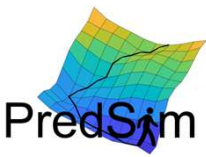
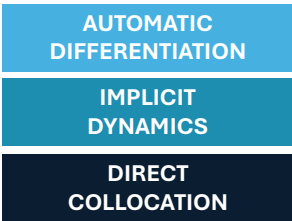
2014

Muscle-driven 3D model (25 dofs, 80 muscles)
~13-17 hours (Lin ... Pandy, 2018)

2018

Muscle-driven 3D model (29 dofs, 92 muscles)
< 1 hour (Falisse ... De Groote, 2019)

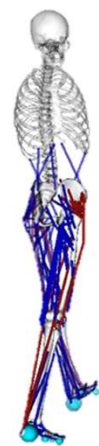
2019



13

13

Predictive simulations of muscle driven 3D walking allow us to explore effect of neuro-musculoskeletal deficits

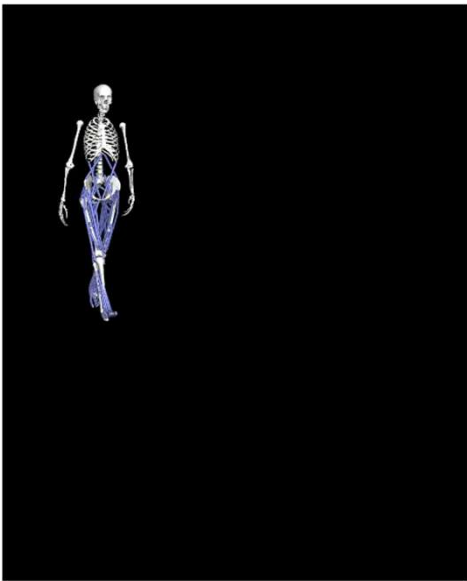


minimize cost



subject to

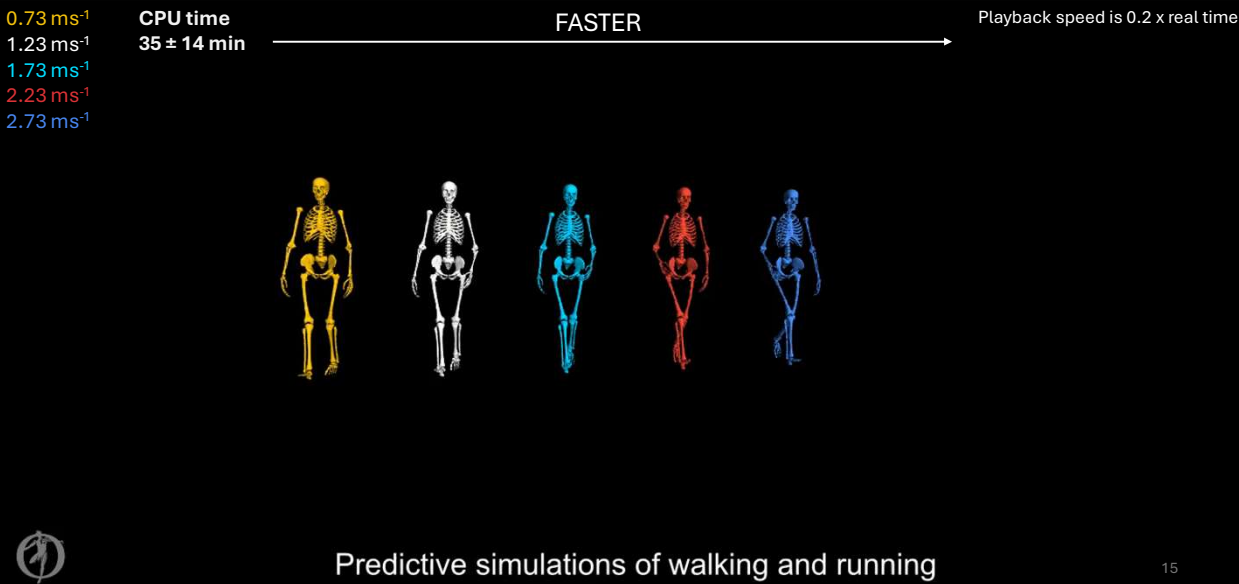
- muscle dynamics
- skeleton dynamics
- symmetry or periodicity
- desired walking speed



Falisse et al., J R Soc Interface, 2019

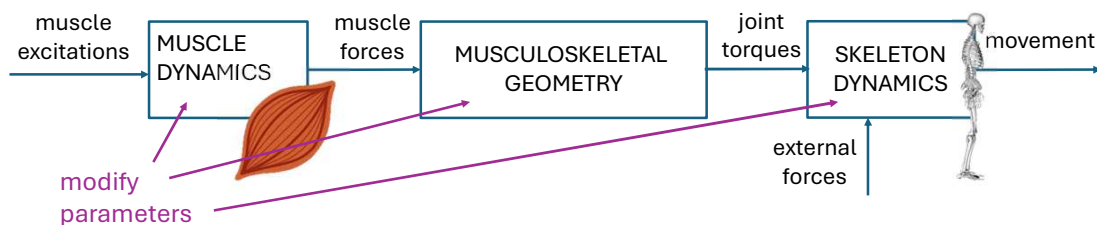
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Model predicts a continuum of walking and running gaits



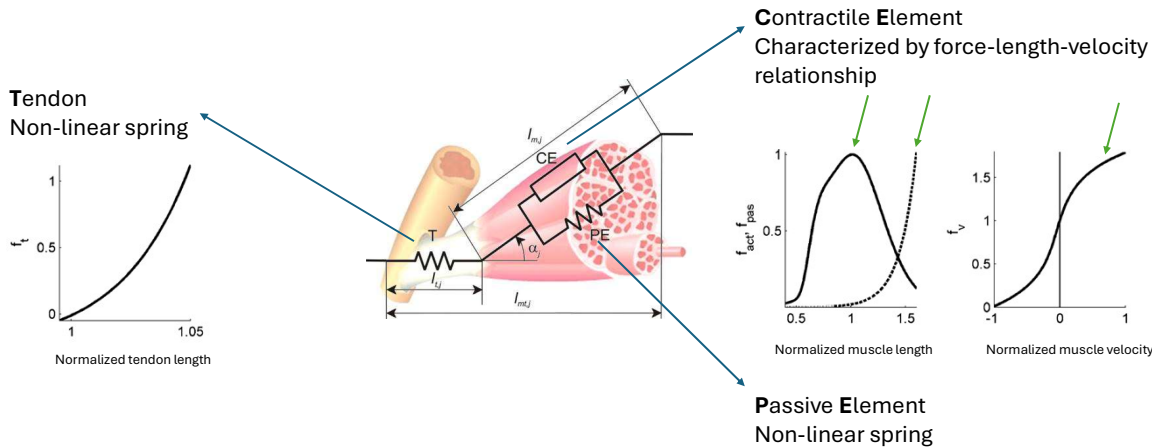
Modeling impairments

minimize cost
with respect to muscle excitations
subject to
dynamic constraints



task constraints, e.g. average forward speed and periodicity for walking

Hill-type muscle model

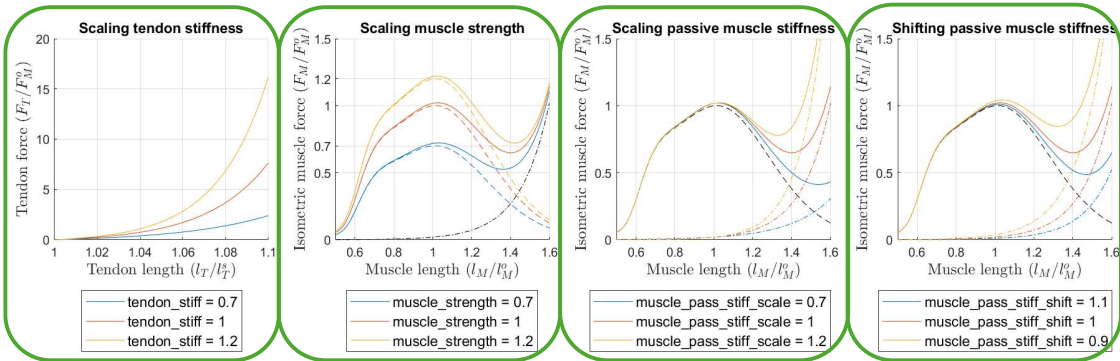


17

Adjusting muscle-tendon properties

Tendon slack length, optimal fiber length, maximal isometric force.

Other properties:



18

Unimpaired
Weakened by 50%
Weakened by 75%
Weakened by 90%

Increasing weakness →

Playback speed is 0.2 x real-time

Effect of weakening the hip muscles on predicted walking patterns

19

Modeling impairments

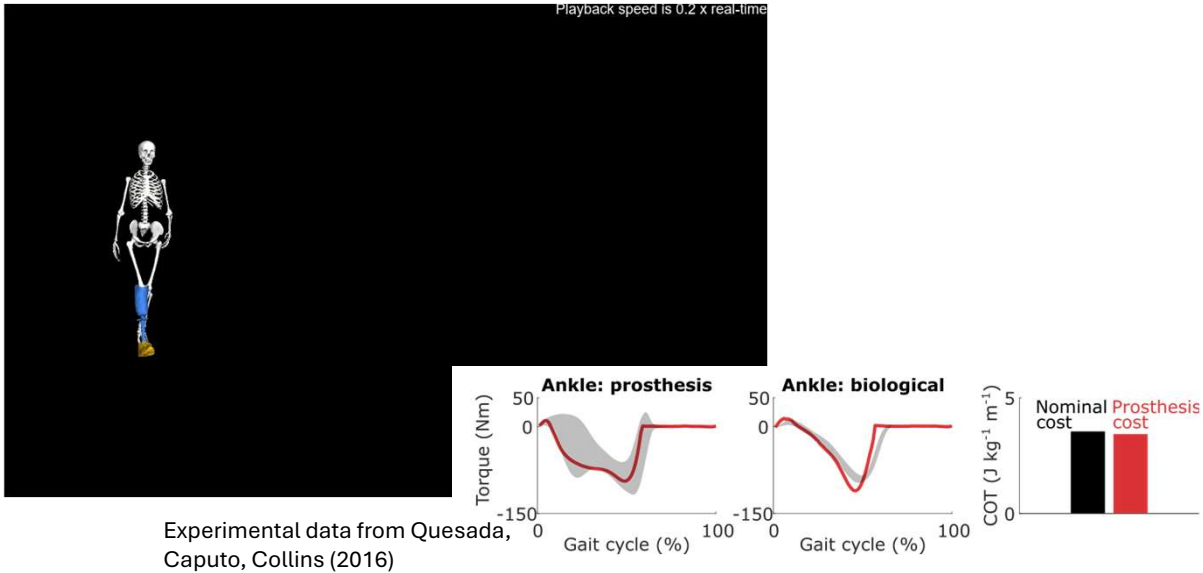
```

graph LR
    A[muscle excitations] --> B[MUSCLE DYNAMICS]
    B -- muscle forces --> C[MUSCULOSKELETAL GEOMETRY]
    C -- joint torques --> D[SKELETON DYNAMICS]
    E[external forces] --> D
    D -- movement --> F[ ]
    style F fill:none,stroke:none
  
```

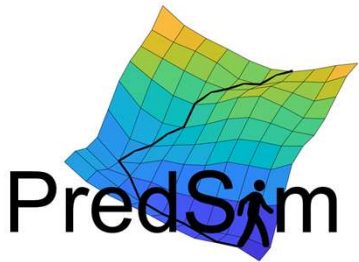
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20

Prosthetic use altered the ankle torque
in agreement with experimental data



21



Skeleton dynamics using modified version of OpenSim that enables algorithmic differentiation



Musculoskeletal geometry

Definition of optimal control problem
Muscle dynamics



CasADi for **algorithmic differentiation and formulating optimization problem** (solved using ipopt)



<https://web.casadi.org/>

22

Why use PredSim software?

PredSim versus OpenSim Moco

	PredSim	OpenSim Moco (Dembia, Bianco et al. <i>PLOS COMP. BIOL.</i> 2020)
Simulation type	Predictive	Predictive, tracking or inverse
Task	Steady-state locomotion	Any task
Discretisation	Direct collocation	Direct collocation
Formulation of dynamics	Implicit	Implicit or explicit
Derivatives	Algorithmic differentiation	Finite differences

AD is faster than FD, and more so for more complex models
(Falisse, Serrancolí et al. *PLOS ONE* 2019)

23

23

Customise and run simulations from a single script

```
% Path to OpenSim model file
osim_path = './Subjects/Falisse_et_al_2022/Falisse_et_al_2022.osim';

% Initialize settings to what was used for this model in the paper
[S] = initializeSettings('Falisse_et_al_2022');

% Path to folder to save the simulation result
S.misc.save_folder = './PredSimResults/Falisse_et_al_2022';

% Select cold-start initial guess
S.solver.IG_selection = 'quasi-random';

% Run simulation
runPredSim(S, osim_path);
```

24

24

OpenSim model file

```
% Path to OpenSim model file
osim_path = './Subjects/Falisse_et_al_2022/Falisse_et_al_2022.osim';
```

Example models:

- | | |
|--------------------------|--------------------|
| • Falisse_et_al_2022 | 31 DoFs 92 muscles |
| • DHondt_et_al_2024_3seg | 31 DoFs 92 muscles |
| • DHondt_et_al_2024_4seg | 33 DoFs 94 muscles |
| • Gait1018 | 10 DoFs 18 muscles |
| • Gait1018_esmac | 10 DoFs 18 muscles |



25

25

A few required settings ...

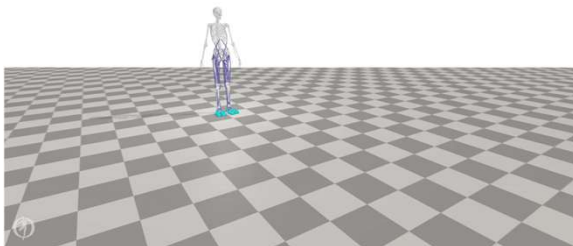
```
% Path to OpenSim model file
osim_path = './Subjects/Falisse_et_al_2022/Falisse_et_al_2022.osim';
```

```
% Initialize settings to what was used for this model in the paper
[S] = initializeSettings('Falisse_et_al_2022');
```

```
% Path to folder to save the simulation result
S.misc.save_folder = './PredSimResults/Falisse_et_al_2022';
```

```
% Select cold-start initial guess
S.solver.IG_selection = 'quasi-random';
```

} important settings: no defaults



Alternative initial guess: OpenSim motion file

- Experimental data
- Previous result
- Custom

26

26

... and many optional settings

- subject
- orthosis
- metabolicEnergy
- weights
- solver
- bounds
- OpenSimADOptions
- post_processing
- misc (task constraints, file paths, scaling, ...)

→ This workshop

```
% Average forward speed of pelvis
S.misc.forward_velocity = 0.8;

% Periodic and symmetric
S.misc.gaitmotion_type = 'HalfGaitCycle';
    initial kinematics, activations
    =
    mirrored final kinematics, activations
    (except forward position)

% Periodic
S.misc.gaitmotion_type = 'FullGaitCycle';
    initial kinematics, activations
    =
    final kinematics, activations
    (except forward position)
```

“initial” can be any point in the cycle
e.g. 80 – 30% is also half cycle

27

27

Today's case studies: Modeling movement disorders and interventions

- Case 1: Muscle weakness and stiffness in Duchenne muscular dystrophy
(Ines Vandekerckhove)
- Case 2: Surgical treatment in cerebral palsy
(Ellis Van Can)
- Casus 3: The effect of an ankle-foot orthosis on drop foot
(Tim van der Zee)

Based on 2D musculoskeletal models.



28

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Workshop: predictive simulations

Case with Duchenne muscular dystrophy (DMD)

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DMD

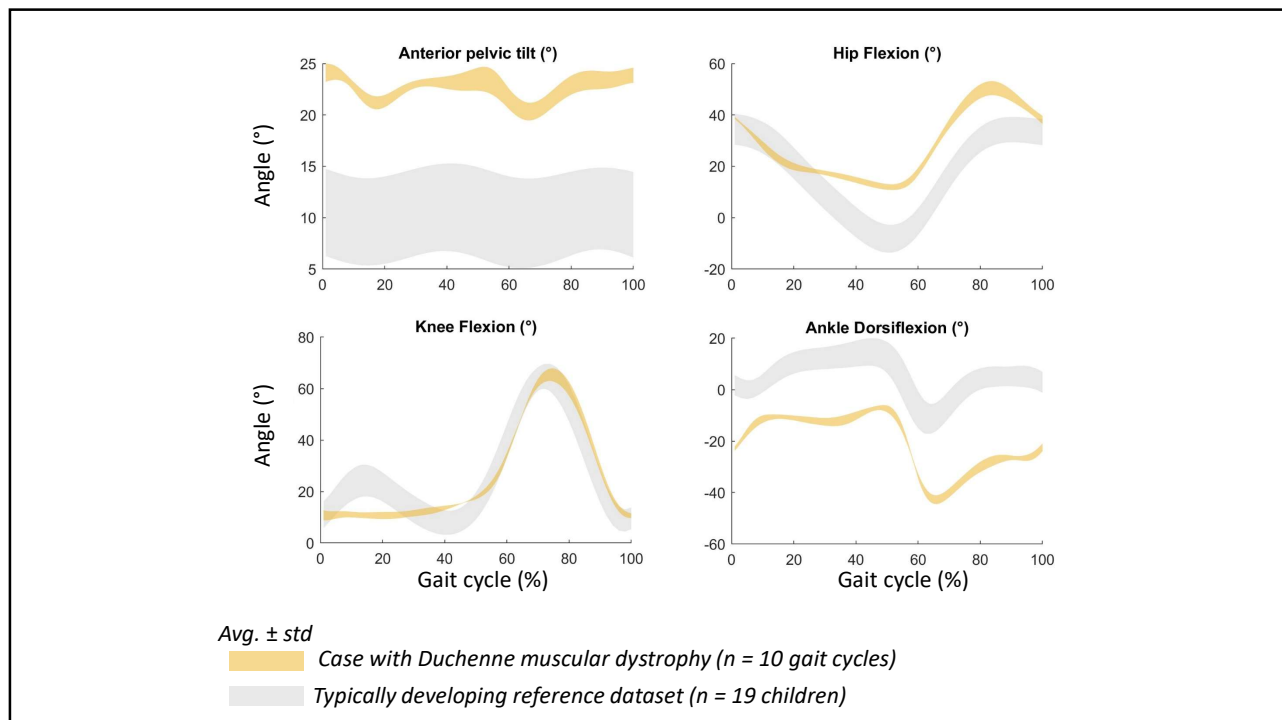
29

Boy with Duchenne muscular dystrophy

15.6 years



30



31

Clinical assessments

Weakness

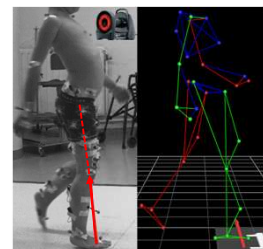
- Fixed dynamometry

Stiffness and contractures

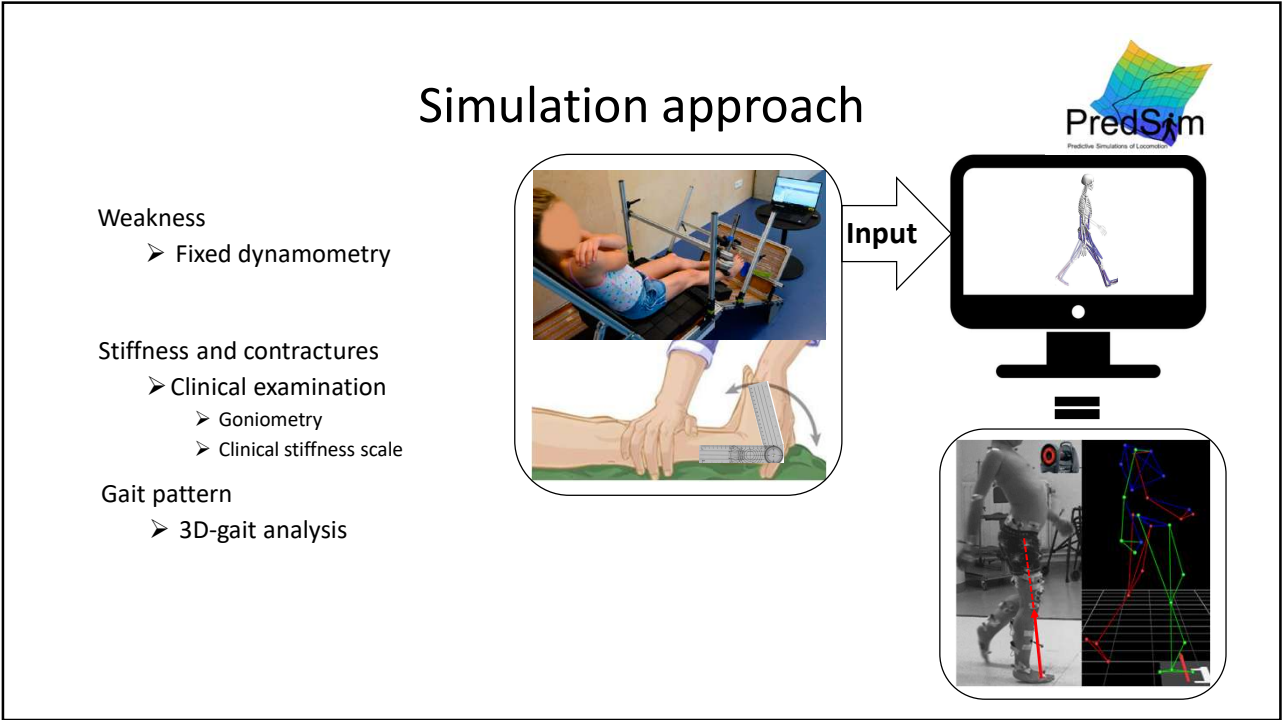
- Clinical examination
 - Goniometry
 - Clinical stiffness scale

Gait pattern

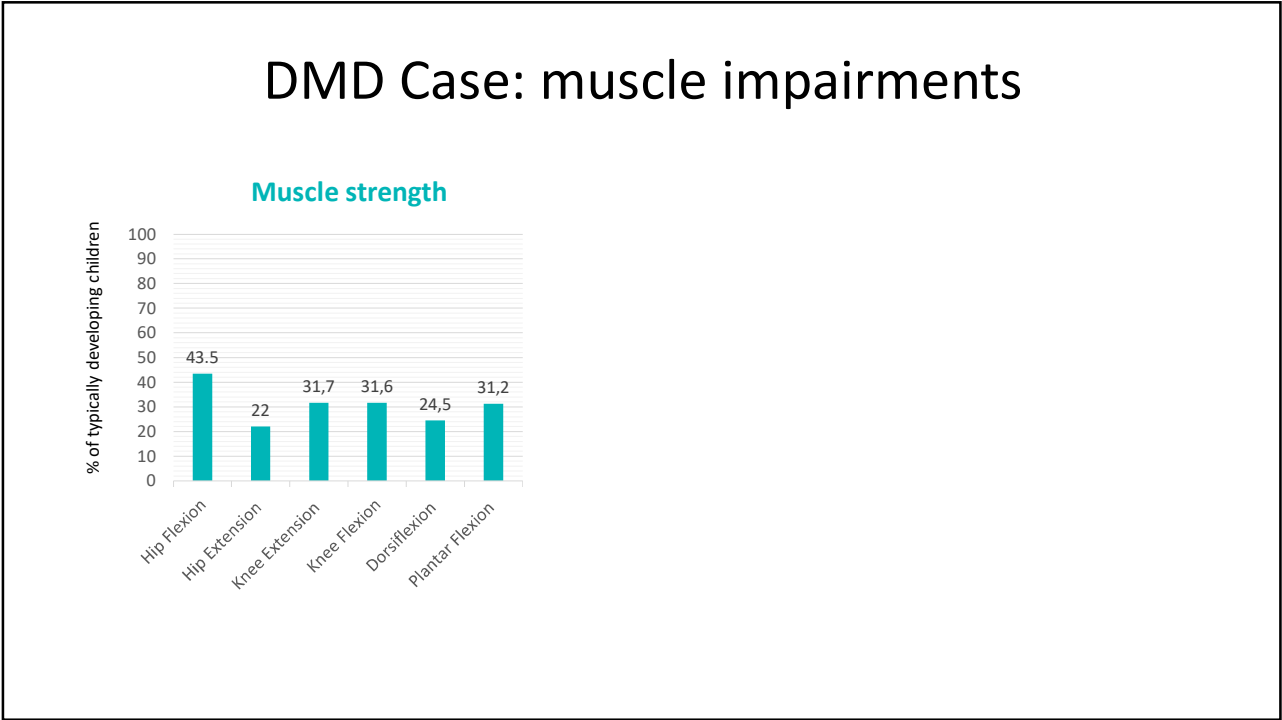
- 3D-gait analysis



32

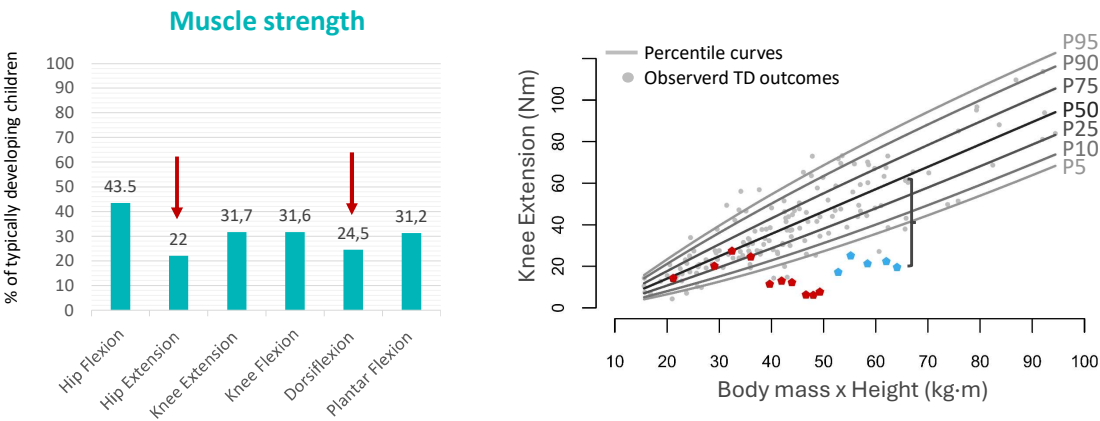


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34

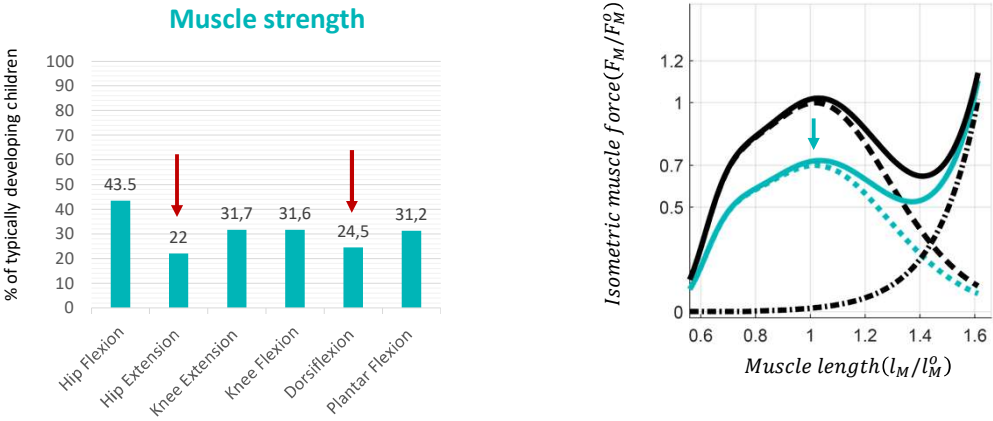
DMD Case: muscle impairments



Vandekerckhove et al., Journal of Anatomy, 2025

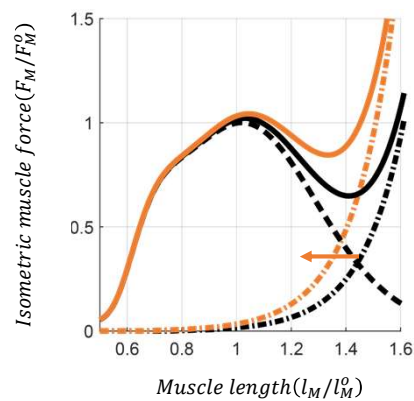
35

DMD Case: muscle impairments



36

DMD Case: muscle impairments



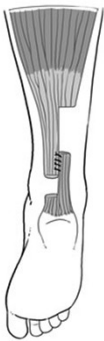
Muscle stiffness/contractures

	Range of motion	Clinical stiffness scale
Modified Thomas test	+5	2
True popliteal angle	-30	1
Silfversköld Test with knee extended	-20	3
Silfversköld Test with knee flexed in 90°	-10	2

37

DMD Case: Achilles tendon lengthening

- Common in the past
- Later controversial
 - Some patients lost ambulation directly after Achilles tendon lengthening
- With predictive simulations based on personalized models, you can test whether such an intervention would improve the gait pattern



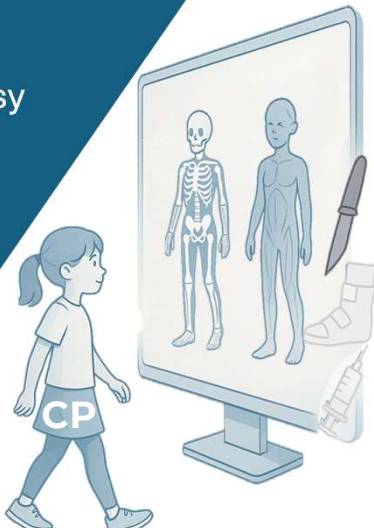
38

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Workshop: predictive simulations

S2: Surgical intervention for a child with Cerebral Palsy

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39

Cerebral Palsy (CP)



Motor control impairments

- Muscle weakness
- Joint hyper-resistance
- Reduced selective control



Musculoskeletal impairments

- Bony deformities
- Contractures

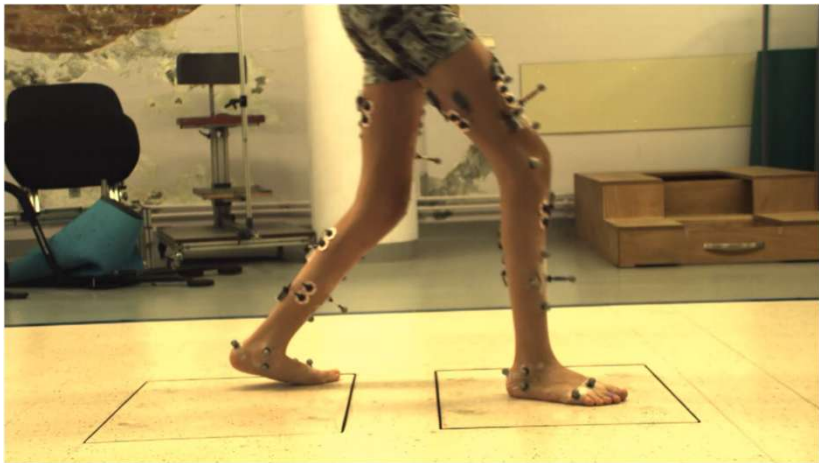


Handsfield et al. (2020)

40

Case CP ESMAC

Spastic diplegic CP, GMFCS II



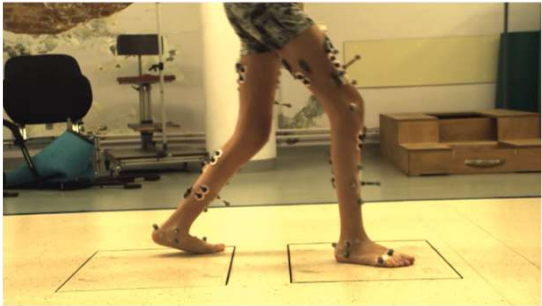
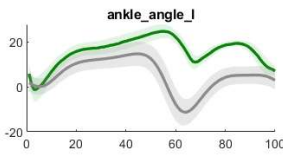
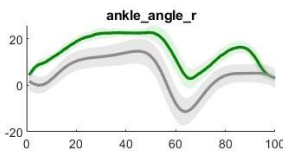
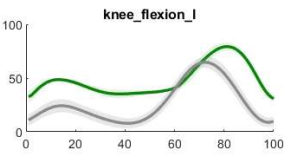
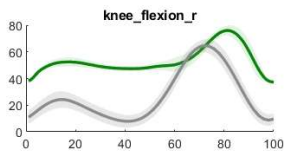
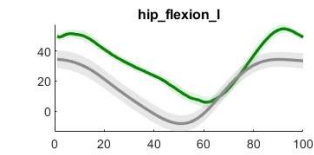
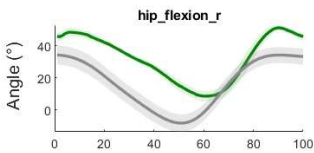
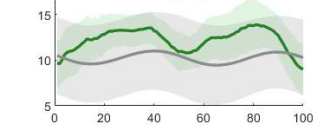
41

Case CP ESMAC

Spastic diplegic CP, GMFCS II

TD reference

pelvis_ant_tilt



CMAL database

42

Case CP ESMAC

Clinical Exam				
	L		R	
Muscle weakness				
	Fl.	Ext.	Fl.	Ext.
Hip				
Knee				
Ankle				
Contractures				
Hamstrings				
Iliopsoas				
Limited Range of Motion				
Knee extension				

43

Case CP ESMAC

Clinical Exam				
	L		R	
Muscle weakness				
	Fl.	Ext.	Fl.	Ext.
Hip				
Knee				
Ankle				
Contractures				
Hamstrings				
Iliopsoas				
Limited Range of Motion				
Knee extension				

44

Case CP ESMAC

Clinical Exam				
	L		R	
Muscle weakness				
	Fl.	Ext.	Fl.	Ext.
Hip				
Knee				
Ankle				
Contractures				
Hamstrings				
Iliopsoas				
Limited Range of Motion				
Knee extension				

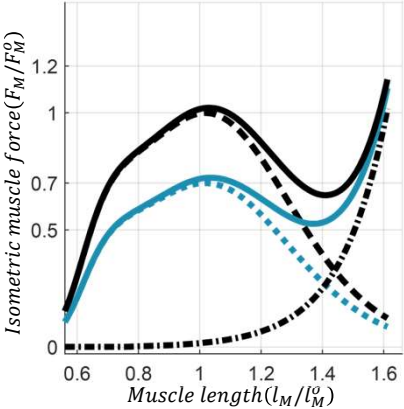
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Case CP ESMAC

Clinical Exam				
	L		R	
Muscle weakness				
	Fl.	Ext.	Fl.	Ext.
Hip				
Knee				
Ankle				
Contractures				
Hamstrings				
Iliopsoas				
Limited Range of Motion				
Knee extension				

- I. Personalize OpenSim model
- Muscle weakness
- Contractures
- Knee extension deficit

Scale factor 1
Scale factor 0.7

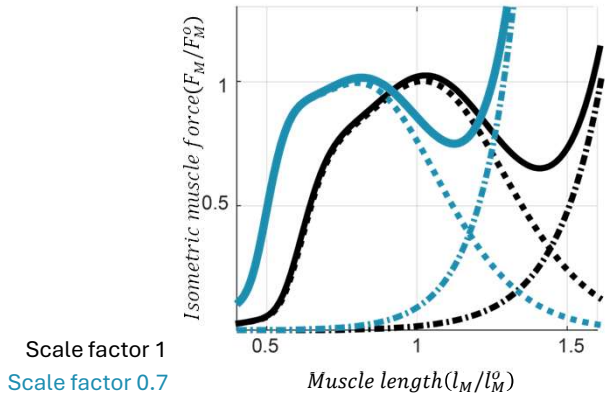


46

Case CP ESMAC

Clinical Exam				
	L		R	
Muscle weakness				
	Fl.	Ext.	Fl.	Ext.
Hip				
Knee				
Ankle				
Contractures				
Hamstrings				
Iliopsoas				
Limited Range of Motion				
Knee extension				

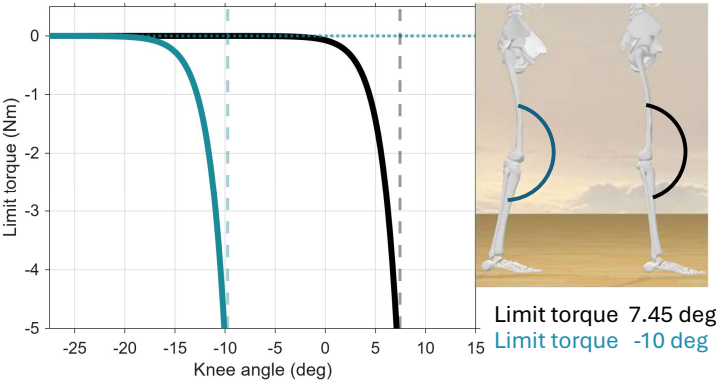
- I. Personalize OpenSim model
- Muscle weakness
- Contractures
- Knee extension deficit



Case CP ESMAC

Clinical Exam				
	L		R	
Muscle weakness				
	Fl.	Ext.	Fl.	Ext.
Hip				
Knee				
Ankle				
Contractures				
Hamstrings				
Iliopsoas				
Limited Range of Motion				
Knee extension				

- I. Personalize OpenSim model
- Muscle weakness
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Case CP ESMAC

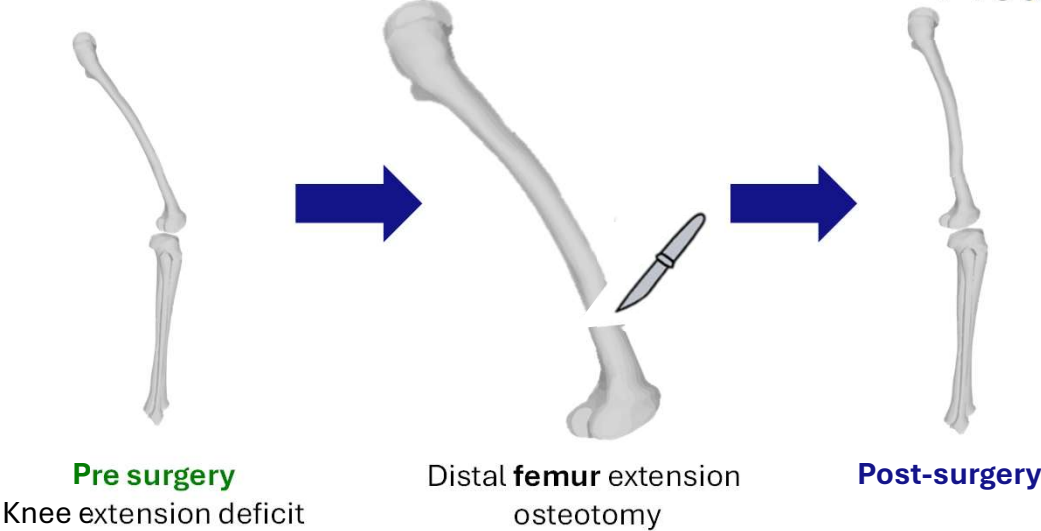
Clinical Exam				
	L		R	
Muscle weakness				
	Fl.	Ext.	Fl.	Ext.
Hip				
Knee				
Ankle				
Contractures				
Hamstrings				
Iliopsoas				
Limited Range of Motion				
Knee extension				

- I. Personalize OpenSim model
- Muscle weakness
- Contractures
- Knee extension deficit

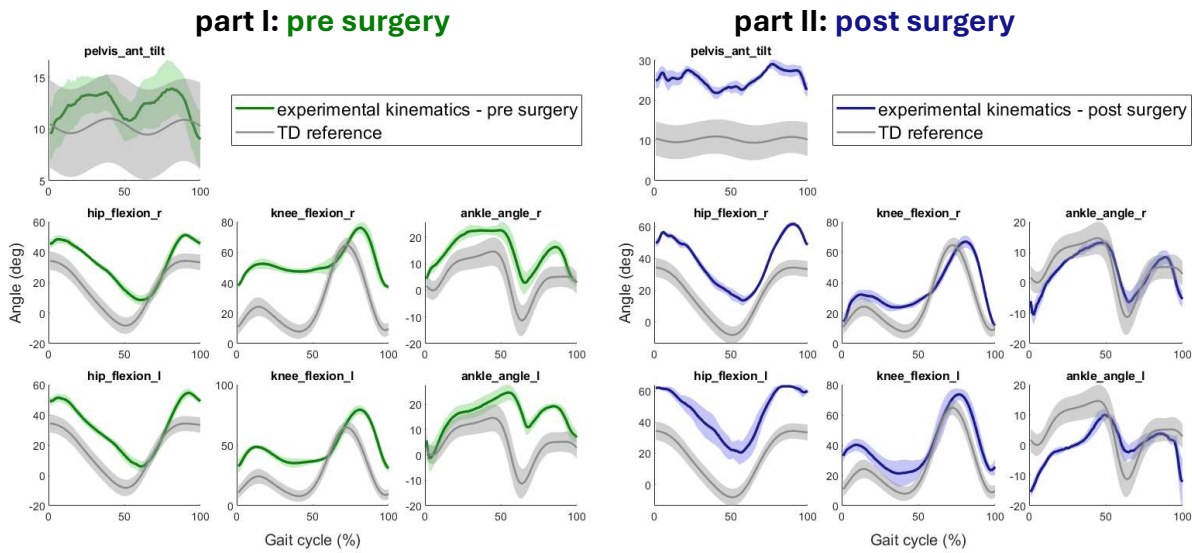
Predict **pre-surgical** gait pattern



Case CP ESMAC – part II: (post) surgery



Compare PredSim results to experimental IK



51

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Workshop: predictive simulations

Case 3: Simulate the effects of an ankle-foot orthosis on gait patterns in individuals with dropfoot

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52

Case 3: effects of an orthosis on drop foot gait

Common cause: nerve damage



53

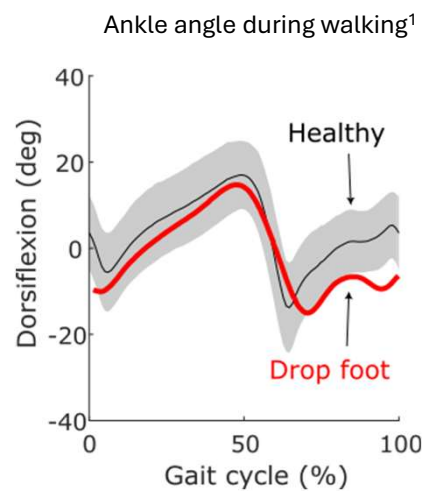
Case 3: effects of an orthosis on drop foot gait

Common cause: nerve damage



54

Case 3: effects of an orthosis on drop foot gait

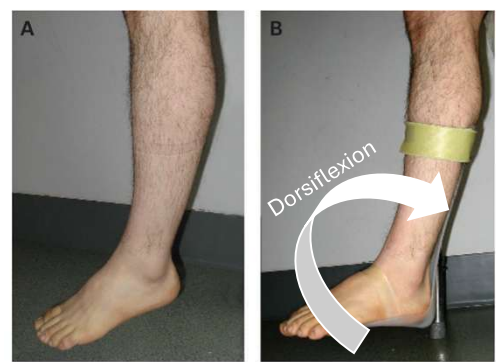


1) Wiszomirska et al., (2017)

Case 3: effects of an orthosis on drop foot gait



Common treatment: ankle-foot orthosis (AFO)



Stewart (2008), *Pract Neurol*, 8: 158-169

Case 3: effects of an orthosis on drop foot gait

Common cause: nerve damage



Other causes:

- Multiple sclerosis
- Stroke
- Cerebral palsy
- Muscular dystrophy

Damage to peroneal nerve
↓
Weakness of tibialis anterior
↓
Inability to lift the foot
↓
Impaired walking

Common treatment: ankle-foot orthosis (AFO)

Device properties:

- Rotational stiffness
- Neutral angle

Unclear which device properties are best

Can we use predictive simulations to inform on AFO design?



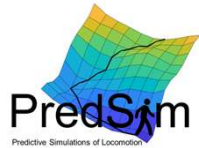
Stewart (2008), *Pract Neurol*, 8: 158-169

How to get started with the workshop

PredSim



<https://github.com/KULeuvenNeuromechanics/PredSim>



Contains the source code to perform movement simulations

Workshop



<https://github.com/KULeuvenNeuromechanics/PredSim-workshop-ESMAC-2026>



Contains code to run workshop simulations via PredSim

Step 1: reference simulation with 2D model

PredSim



Workshop



Running a reference 2D simulation with PredSim

Before going to the hands-on tutorials, the user should run a reference 2D simulation with [PredSim](#).

- 1. Open Matlab
- 2. Navigate to your `PredSim` folder in Matlab
- 3. Open the script `main.m` in Matlab by clicking on it
- 4. During this workshop the user will run predictive simulations with the 2D model instead of the default 3D model. Therefore, adjust the following lines in `main.m`:

o Line 20:

```
[S] = initializeSettings('gait1018_esmac');
```



o Line 25:

```
`S.subject.name = 'gait1018_esmac';
```



o Line 51-52:

```
result_paths{1} = fullfile(pathRepo,'Tests','ReferenceResults',...  
    'gait1018_esmac','gait1018_esmac_reference.mat');
```



- 5. Click on the green 'Run' button

Step 1: reference simulation with 2D model

PredSim



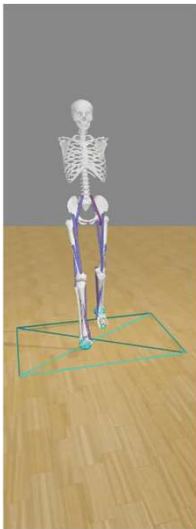
Workshop



Visualizing in OpenSim

To visualize the simulation in OpenSim:

- 1. Open OpenSim
- 2. Click on 'File > Open Model...' and navigate to the 2D model
`PredSim/Subjects/gait1018_esmac/gait1018_esmac.osim`
- 3. Click on 'File > Load Motion...' and navigate to the mot file of your simulation
`PredSimResults/gait1018_esmac/gait1018_esmac_v1.mot`
- 4. Click on the 'Play forward' button to see the motion. You may also adapt the playback speed.



How to get started with the workshop

PredSim



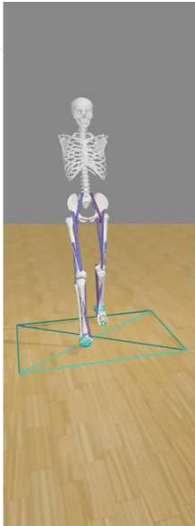
Getting started with one of the cases

Before starting one of the three cases, make sure that this repository is added to your Matlab path.

- 1. Either download or clone [the current repository](#)
- 2. Open Matlab
- 3. Navigate to the `PredSim-workshop-ESMAC-2026` folder
- 4. Open the script called `set_up_paths.m`
- 5. Click on the green 'Run' button

Now you're ready to start with one the three cases!

- [S1 DMD](#)
- [S2 CP](#)
- [S3 Orthosis](#)



Workshop

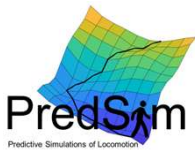


How to get started with the workshop

PredSim



<https://github.com/KULeuvenNeuromechanics/PredSim>



Contains the source code to perform movement simulations

Workshop



<https://github.com/KULeuvenNeuromechanics/PredSim-workshop-ESMAC-2026>

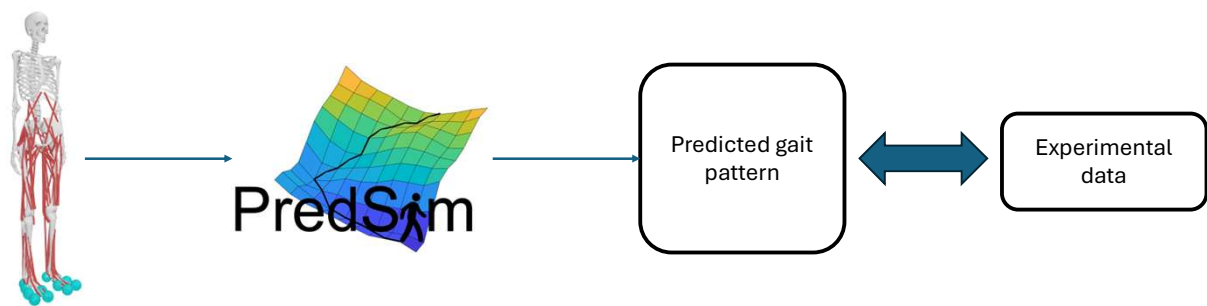


Contains code to run workshop simulations via PredSim

Predictive simulations in 3D

63

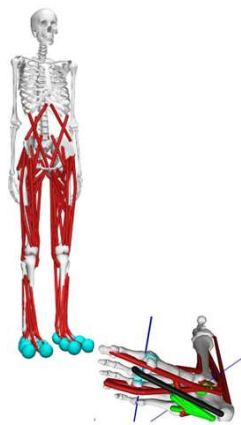
Validation



64

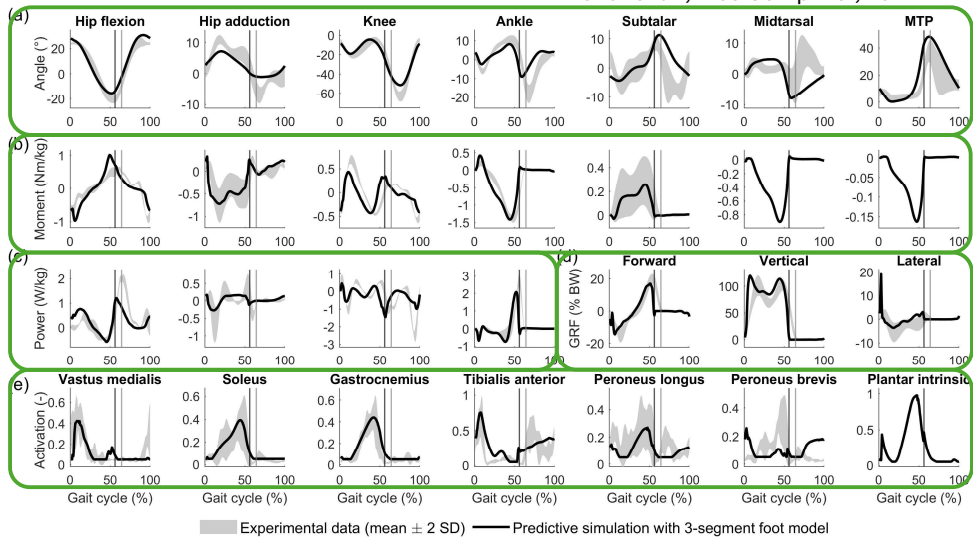
64

Simulations based on DHondt_et_al_2024 capture walking pattern of unimpaired adult



Extended version of OpenSim's gait2392
33 dofs
94 muscles

D'Hondt et al., Plos Comp Biol, 2024

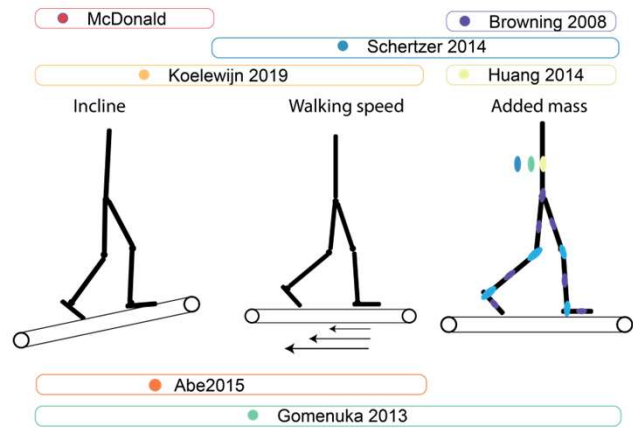


65

Do optimal control simulations predict gait mechanics and energetics across walking conditions?



Maarten Afschrift

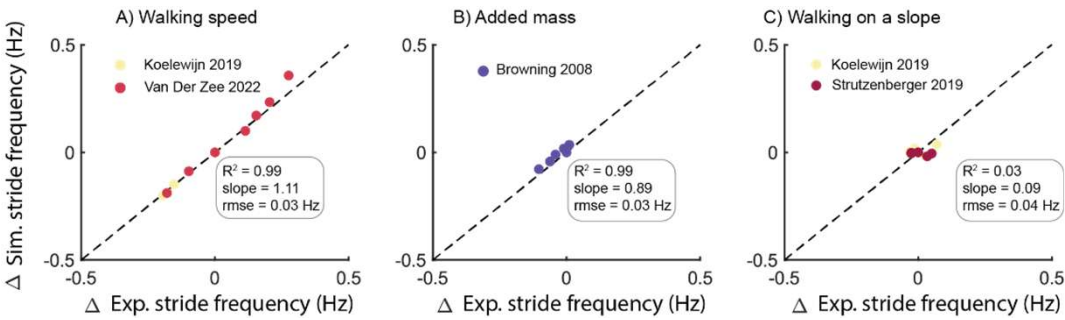


Falisse et al., Plos One, 2022; Afschrift et al., PLOS Comp Biol, 2025

66

66

Simulations capture stride frequency across conditions

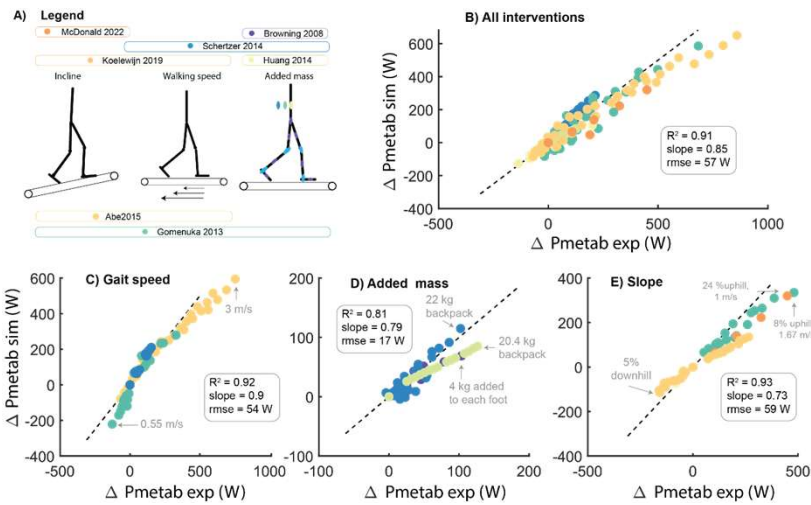


Afschrift et al., *PLOS Comp Biol*, 2025

67

67

Optimal control simulations capture main trends but not fine details of healthy gait energetics



Afschrift et al., *PLOS Comp Biol*, 2025

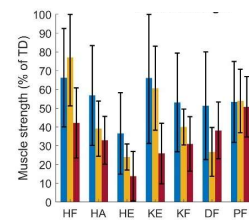
68

Overall, predictions of mechanics/ activations are better than predictions of metabolic energy.

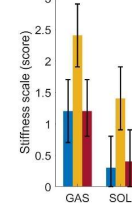
68

Contractures in Duchenne Muscular Dystrophy
- Should we treat them or are they a useful compensation strategy?

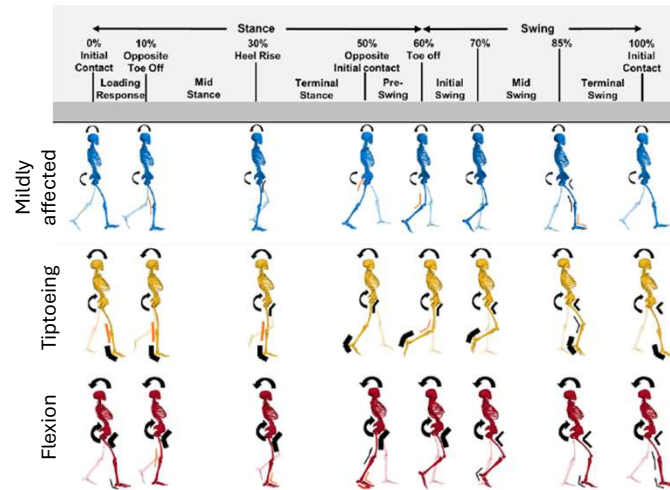
WEAKNESS



CONTRACTURES



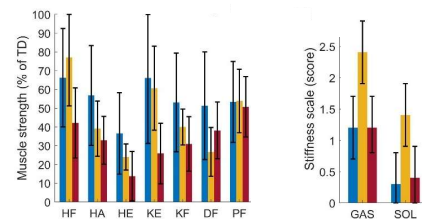
Walking patterns based on a large longitudinal (30 boys – 137 measurements)



Vandekerckhove et al., Scientific Reports, 2024

69

Contractures in Duchenne Muscular Dystrophy
- Should we treat them or are they a useful compensation strategy?



WEAKNESS ONLY



Tiptoeing in the absence of contractures!

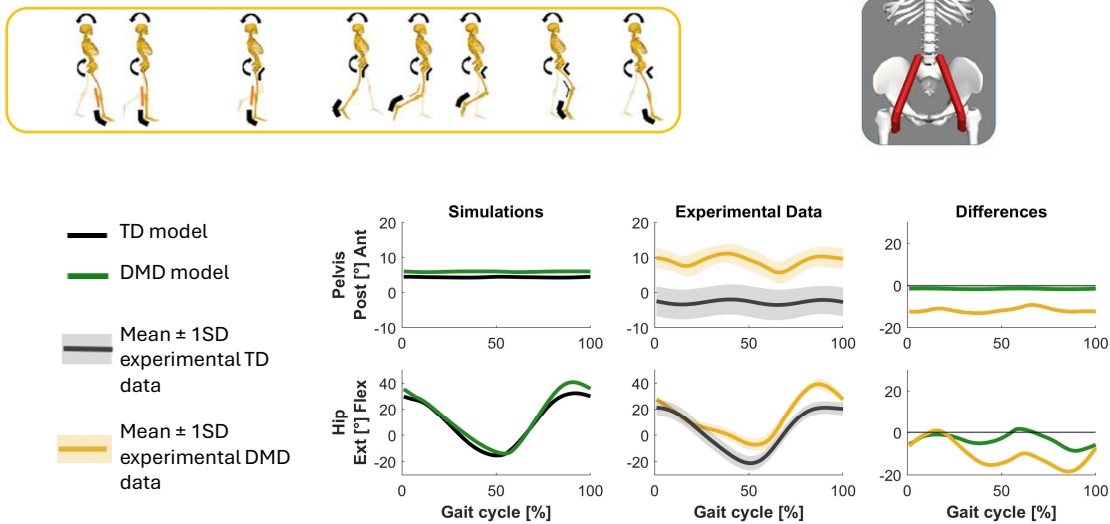
WEAKNESS AND CONTRACTURES



Vandekerckhove et al., JNER, 2025

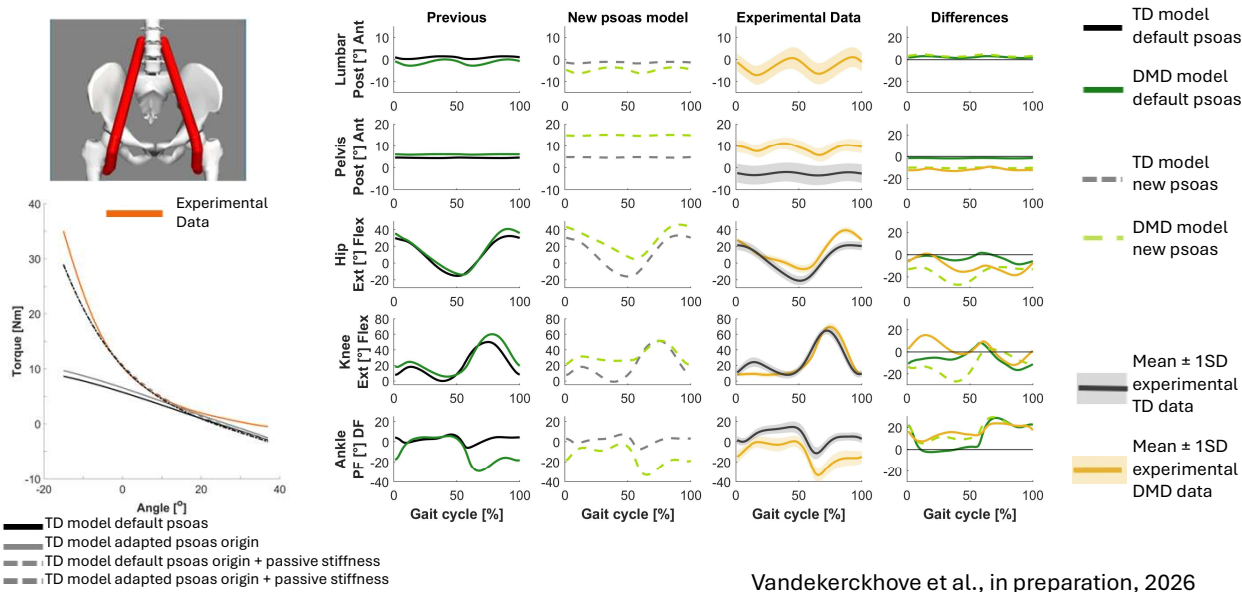
70

Simplified psoas model does **not** capture increased anterior pelvic tilt and hip flexion



71

New psoas model does **capture** increased anterior pelvic tilt and hip flexion
- Adapted psoas origin to lumbar spine and passive psoas stiffness matched to experimental data



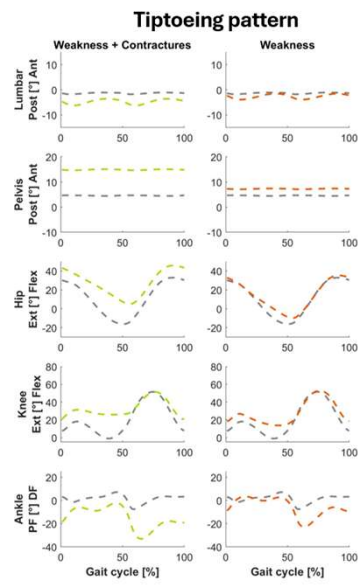
Vandekerckhove et al., in preparation, 2026

72

Revisiting our initial clinical question with the new psoas model
- Should we treat contractures or are they a useful compensation strategy?

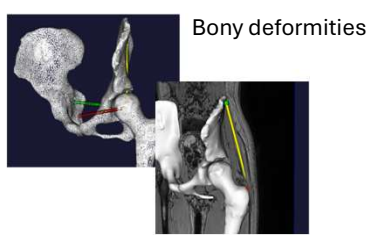
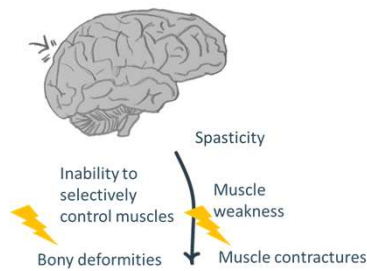
Simulations suggest:

Yes, for hip and knee flexion contractures
But tiptoeing pattern remains



73

What is the contribution of musculoskeletal deficits to alterations in gait kinematics in cerebral palsy?

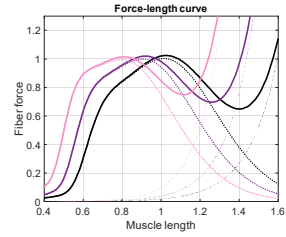
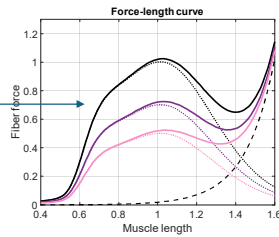


Bram Van Den Bosch

CONTRACTURES - Reduced optimal fiber length such that resistance torque at end of range of motion is 15Nm

Gait and balance impairments

WEAKNESS - Reduced maximal muscle force based on manual muscle testing score

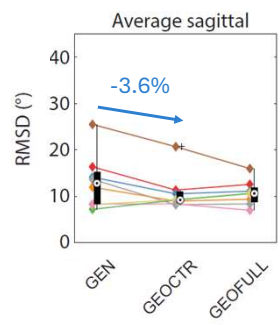
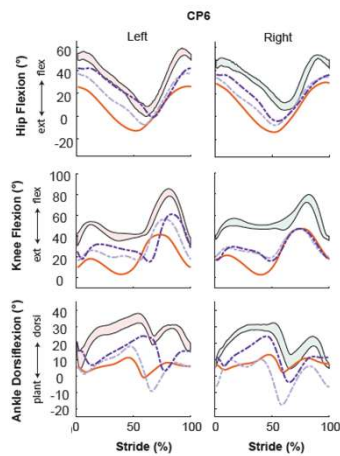


Van Den Bosch et al., *JNER*, 2025

74

74

Modest and variable contributions of musculoskeletal impairments to alterations in gait kinematics in cerebral palsy



- In line with modest and variable improvements after multi-level surgery
- Remaining differences between simulated and experimental gait kinematics suggest large contribution of motor control impairments

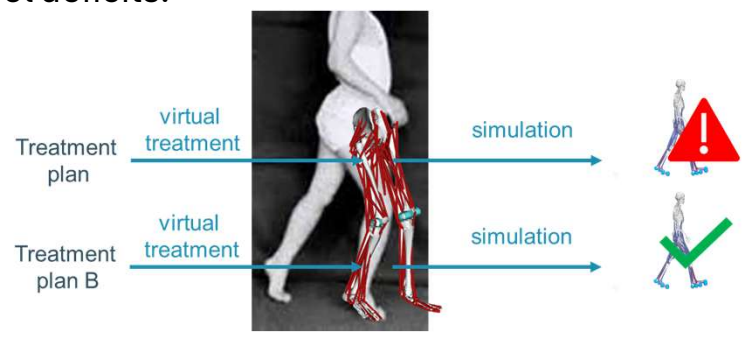
Left experimental ± SD
Right experimental ± SD
Scaled generic model = GEN
Bony deformities + contractures = GEOCTR
Bony deformities + contractures + weakness = GEOFULL

Van Den Bosch et al., JNER, 2025

We still need to demonstrate benefits of using predictive simulations for treatment selection and design

Challenges

- Improve accuracy of simulations.
- Model motor control deficits.



Opportunities and barriers for clinical implementation of predictive simulations



[PollEv.com/tombuurke748](https://pollev.com/tombuurke748)

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77

77

Where do you see the most important need or application for predictive simulations in clinical practice?

Nobody has responded yet.
Hang tight! Responses are coming in.

Start the presentation to see live content. For screen share software, share the entire screen. Get help at pollev.com/app

78

Which barriers need to be overcome before clinical implementation? (And how?)

Nobody has responded yet.
Hang tight! Responses are coming in.

Start the presentation to see live content. For screen share software, share the entire screen. Get help at pollev.com/app

79

Our goals for today

Now you...

- know the theory underlying musculoskeletal modelling and predictive simulations.
- can perform predictive simulations with 2D models in *PredSim*.
- can find information and sources for 3D simulations.
- know the opportunities and barriers for clinical implementation of predictive simulations.

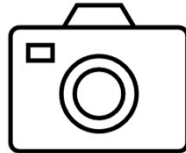


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80

80

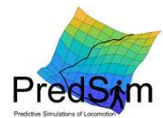
Group photo



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81

81



Predictive simulations of pathological gait: from clinical data to digital twins

Tom Buurke, Friedl De Groote, Ines Vandekerckhove, Tim van der Zee, Ellis Van Can,
Míriam Febrer Nafria, Lars D'Hondt, Stefanie de Jager

Workshop W1 ESMAC 2026, Maastricht, NL

umcg:

KU LEUVEN



82